REFERENCE · R1

VEHICLE REFERENCE CARD

Vehicle specs, fluids, pressures & torque values at a glance

2009 LC76 SW · 1HZ 4.2L Diesel · low-boost turbo (0.6 bar) · intercooler
Last updated: 12 October 2026

VEHICLE IDENTITY

▸ IDENTIFICATION NUMBERS

VIN	JTEEB71J207007860
Engine No.	1HZ0633958
Registration	CAA 430 689
MM Code	60078101
Model Code	HZJ76R-RKMRS
Year	2009
Colour	Beige 4E9 / Gray FG10

▸ ENGINE SPECIFICATIONS

Engine	1HZ · 4164cc · 6-Cyl
Bore × Stroke	94mm × 100mm
Compression	22.4:1
Injection	Indirect · Swirl Chamber
Cam Drive	Rubber Timing Belt
EGR	None
Management	Fully Mechanical

▸ POWER — STOCK VS. TUNED

Stock Output	96 kW / 285 Nm
Low Boost Turbo	0.6 bar · 118 kW / 373 Nm
Fitted By	SAC in 2011 @ 38,429 km
Intercooler	SAC fitted 02/2012
Snorkel	Safari Snorkel

DIMENSIONS & GEOMETRY

▸ BODY DIMENSIONS

Wheelbase	2,730 mm
Overall Length	4,890 mm
Overall Width	1,870 mm
Overall Height	~1,920 mm (w/ rack)
Front Track	1,520 mm
Rear Track	1,425 mm (95mm narrower)

▸ OFF-ROAD GEOMETRY

Ground Clearance	~275 mm (w/ 2" lift)
Angles	Not measured — break-over is the tightest (long wheelbase)
Wading Depth	~800 mm (w/ snorkel)
Lift	OME 40–50 mm (2")

▸ WEIGHTS & CAPACITY

Kerb Weight	2,190 kg (factory)
GVM	3,000 kg
Payload	~683 kg (as-built est.)
Tow (braked)	3,500 kg
Tow (unbraked)	750 kg
Fuel Tank	Replacement 145 L

FLUIDS & CAPACITIES

▸ ENGINE BAY FLUIDS · OWNER RECOMMENDATION

ENGINE OIL
Shell Rimula R4 X 15W-40
11.3 L (with filter) — buy 12L for change + top-up reserve

COOLANT
Toyota Long Life Coolant
10.9 L · 50/50 distilled water

BRAKE & CLUTCH FLUID
DOT 4
Shared reservoir · top-up only

POWER STEERING
ATF Dexron II
1.3 L

▸ DRIVETRAIN FLUIDS · OWNER RECOMMENDATION

GEARBOX R151F
Castrol Transmax Manual 75W-90
2.2 L

TRANSFER CASE
Castrol Transmax Manual 75W-90
2.1 L

FRONT & REAR DIFFERENTIALS
Castrol Transmax Axle EPX 85W-140
Front 2.6 L · Rear 2.4 L

SUSPENSION, WHEELS & TYRES

▸ SUSPENSION — OLD MAN EMU

System	OME Medium · 400 kg constant
Fitted By	4x4 Megaworld
Lift	40–50 mm (2 inch)
Front Springs	OME Coil 859
Rear Springs	OME Leaf CS058R

▸ FIELD TORQUE & PRESSURES

Wheel Nuts	131 Nm · 16" steel (VERIFY vs handbook — OI-7)
Re-torque	After any wheel change — re-check at ~50 km

▸ WHEELS & TYRES

Tyre Size	265/75R16 (LRD 119/116 S)
Tyre Brand	BFG All-Terrain KO3
Rims	Standard OEM 16"
TPMS	Steel Mate Solar (internal)

Front Shocks	OME Nitrocharger+ N181
Rear Shocks	OME Nitrocharger+ 63111

Tyre Pressures	Terrain table authoritative in R14
Wheel Procedure	Change & bead work: R40
Other Torques	Engine/drivetrain: R4 · workshop manual

▸ DRIVETRAIN

Gearbox	R151F · 5-Speed Manual
5th Gear	Terrain Tamer — Replacement
Clutch	Terrain Tamer — 4WD Replacement
4WD	Part-Time · 2H / 4H / 4L
Front Hubs	Auto + Manual Lock

ELECTRICAL SYSTEM & CAMP SETUP

▸ CAMP SETUP

Roof Rack	African Out + Top Rail + Ladder
Table	Aluminium under Roof Rack
Storage	Double Drawer + Fridge
Seat Covers	Canvas — Escape
Cooking	5L Gas + Twin Top Cast Iron Cooker

▸ DUAL BATTERY & POWER SYSTEM

Crank Battery	Enertec 650B-BLS · 92Ah · 750 CCA
House Battery	Enertec 210Ah LiFePO4
DC-DC	Victron Orion XS 12/12-50A · ~30A cap
Monitor	Victron BMV-712 Smart
Inverter	Victron MultiPlus 12/500 · ~430W

RECOVERY, ARMOUR & ACCESSORIES

▸ FITTED RECOVERY

Winch	T-Max 9500 lbs
Bull Bar	ARB Bumper
Skid Plate	Mid Skid Plate
Tow Bar	H/D Gooseneck Detach
Recovery Boards	2× MaxTrax
Compressor	ARB vehicle mounted

▸ RECOVERY GEAR

Shackles	Metal & soft shackles
Tow Rope	Standard tow rope
Snatch Rope	Kinetic recovery rope
Tree Protector	Winch tree saver
Pulley Block	Snatch block
Line Dampener	Winch line dampener
Spade	Recovery spade
PPE	Leather gloves · safety glasses
Jack	Hydraulic & bottle jack

▸ OTHER ACCESSORIES

Compressor	T-Max portable
Mains Charging	Victron MultiPlus 12/500 via 16A shore inlet
Fuel Canisters	2× Desert Fox 20L
Spotlights	IPF 800 Extreme
Solar Panels	Hardkorr 200W Portable
Tyre Inflation	Indeflate 2-Hose Unit
Jump Start	ARB Portable lithium jump pack
Jump Start	Heavy duty jumper cables
Seed Net	SecureTech Radiator · 800×1000 mm

KNOWN ISSUES

Rear Track Width Rear track ~95–100 mm narrower than front. Options: -55 offset rear rims, wheel spacers, or TrueTrac correction.	**Castor Correction** Incorrectly set by Johan Tyre. Corrected by Conrad at Suspension City, Maitland.	**5th Gear Weakness** OEM 5th gear known failure point. Terrain Tamer HD fitted. Avoid low-RPM in 5th — change to 4th until at cruising speed.
Head Gasket (1HZ) Never overheat. White exhaust smoke = coolant in combustion. Stop immediately. Check level every morning.	**Valve Clearance & Pump Timing** Critical for performance on 1HZ. Requires specialist — half a degree on pump timing makes a significant difference.	**Terrain Tamer 5th Gear Upgrade** 0.711:1 reduction ratio kit available. Reduces RPM at cruise, improves fuel economy vs current OEM-ratio 0.838:1.